Elijah Kwaku Adutwum Boateng

LinkedIn: Elijah Kwaku Adutwum Boateng
Address: Frimpongwe Street 4, North Kaneshie - Accra
Email: boateng.elijah@ashesi.edu.gh
Github: github.com/eb-adutwum
Mobile: +233-244-870-609

Education

Ashesi University — Berekuso, Ghana
MPhil. Intelligent Computing Systems — *2025 - 2027*

Ashesi University — Berekuso, Ghana
BSc. Computer Science — *2019 - 2023*

Honors and Awards

Best Computer Science Thesis Award, Ashesi University	Jun 2023
1^{st} Place, Career Fair Programming Challenge (Freshman & Senior year), Ashesi University	Mar 2020, Mar 2023
Silver Medallist, 2022 IGEM Competition, IGEM Foundation	Oct 2022
Grant Recipient, IGEM Impact Grant (Team AshesiGhana), IGEM Foundation	Jul 2022
Grant Recipient, Safety and Biosecurity Grant (Team AshesiGhana), IGEM Foundation	Jul 2022
MasterCard Foundation Scholar	Sep 2019 - Jun 2023

Research Projects

Interius: Autonomous Backend Engineering System — Accra, Ghana
Research Engineer – Multi-Agent Systems — *2026*

- **Project**: Designed and implemented an autonomous system that transforms natural-language specifications into structured, multi-file backend applications through a staged multi-agent pipeline with runtime validation.
- **Project Scale and Contributions**:
  - Architected a multi-stage generation pipeline comprising requirements synthesis, architecture design, implementation, review, and repair, enforcing structured artifact flow across stages.
  - Developed specialized agents with distinct responsibilities, enabling decomposition of backend generation into modular, inspectable reasoning steps.
  - Integrated retrieval-augmented reasoning and artifact persistence to maintain coherence across multi-file code generation.
  - Implemented sandbox-based runtime validation and automated repair loops to detect and fix execution failures before deployment.
- **Impact**:
  - Achieved 90% end-to-end pipeline success rate and 85% first-attempt generation success on backend tasks.
  - Improved reliability and recoverability of backend generation compared to single-pass LLM approaches through structured validation and repair.

  Project Type: Research and Systems Engineering

Automatic Speech Recognition — Berekuso, Ghana
Research Engineer – Applied DL Project — *Nov 2025*

- **Project**: Fine-tuned a pretrained Whisper-Tiny Transformer model for automatic speech recognition under extreme data scarcity, focusing on Ghanaian-accented (Twi-influenced) English speech from the AfriSpeech dataset.
- **Project Scale and Contributions**:
  - Worked with a low-resource AfriSpeech subset comprising 1,315 audio–text pairs with significant accent and pronunciation variability.
  - Fine-tuned the Whisper-Tiny model using transfer learning and evaluated performance using Word Error Rate (WER) and Character Error Rate (CER).
  - Benchmarked performance against earlier RNN–CTC baselines to quantify improvements from pretrained Transformer-based speech models.
- **Results**:
  - Reduced ASR error from near-random RNN–CTC performance (approximately 85% WER) to 34.28% WER and 19.07% CER.
  - Demonstrated the effectiveness of pretrained Transformer models for speech recognition in low-resource African language contexts.

  Project Type: Independent Project

Ashesi University, CSIS Department Berekuso, Ghana
Research Assistant — AI-Based Evaluation of Handwritten Assessments for Fairness *Sep 2024 – Aug 2025*

- **AI-driven automated grading system**: Designed and implemented an end-to-end grading pipeline integrating OCR and large language model APIs (OpenAI, Anthropic, DeepSeek, Grok) to evaluate handwritten and descriptive exam scripts with a focus on consistency, fairness, and ethical compliance.
- **Project Scale and Contributions**:
  - Processed and analyzed over 1,200 scanned handwritten scripts across Algorithms, Data Structures, Artificial Intelligence, Calculus, and Discrete Structures courses.
  - Designed and conducted faculty and student surveys and focus groups to evaluate perceptions of transparency, trust, and fairness in AI-assisted grading.
  - Performed simulation-based evaluations and experimental testing to study accuracy trends, robustness, and stability under handwriting variability.
- **Impact**:
  - Demonstrated measurable improvements in grading consistency and reduced bias across diverse handwriting styles.
  - Informed institutional discussions on potential integration with the Canvas Learning Management System.
  - Contributed to the initial draft of a conference paper exploring automated assessment and academic fairness.

Supervisors: Prof. Sampson Asare, Dr. Govindha Yeluripati, Dr. Ayorkor Korsah

The Aurum Institute Accra, Ghana
AI Engineer *Apr 2024 - Jun 2024*

- **Developed an AI-driven treatment recommendation system**: Developed a Retrieval-Augmented Generation (RAG) model for recommending treatments based on Ghana's Standard Treatment Guidelines (STGs) for infectious diseases.
- **Project Scale**:
  - Focused on 40 high-priority infectious diseases, including Bacterial Meningitis, Tuberculosis, and Sepsis.
  - Utilized a dataset of 50,000 anonymized patient records and 1,000 STGs from the Ministry of Health.
  - Integrated with some hospital management systems for experimental testing.
- **Impact**:
  - Increased treatment adherence by more than 20% through personalized, context-aware recommendations.
  - Reduced recommendation time from 7 to 2 minutes, enhancing doctors' efficiency up to 35%.
  - Enabled about 92% compliance with the latest national guidelines through regular protocol updates.

Supervisor: Nana Kofi Quakyi, MPH

Ashesi University, University of Ghana Berekuso, Ghana
Research Assistant – Multi-Constrained Face Recognition *Summer 2023, Jul 2021 – Dec 2022*

- **Project**:
  - Investigated novel approaches to face recognition under multiple constraints, focusing on occlusion recovery, low-resolution image enhancement, and computational efficiency.
  - Implemented and optimized a hybrid DWT-PCA/FFT algorithm.
- **Contributions**:
  - Conducted comprehensive literature review on multi-constrained face recognition techniques, identifying key research gaps and potential areas for improvement and innovation.
  - Run experiments under single and combined constraint settings.
  - Developed an interface program to seamlessly integrate MATLAB and Python code bases, enhancing the project's overall efficiency and enabling more complex analyses.
  - Created a simulation program to model various occlusion scenarios, allowing for robust testing of the face recognition algorithm under diverse real-world conditions.
  - Created infographics for papers and presentations.
- **Impact**:
  - Demonstrated that the combination of wavelet transforms and PCA feature extraction significantly outperforms standalone methods in multi-constrained environments.
  - Wavelet-based methods improved recognition accuracy by 15% in occluded face scenarios.
  - This research laid the foundation for the publication in the Scientific African Journal and my award-winning undergraduate thesis.

Advisors: Dr. Joseph A. Mensah & Dr. Justice K. Appati

International Genetically Engineered Machine (IGEM) Competition Paris, France
Silver Medallist, Team Ashesi – Ghana *Oct 2022*

- **Project**: Developed a multi-functional biosensor by engineering several E. coli cultures that either detect the presence of gold or some of the selected pathfinder elements (Arsenic and Iron).
- **Contributions**:
  - Conducted core research on biosensor structure, identifying chromoproteins as a cost-effective and practical alternative to fluorescent proteins for visual detection.
  - Collaborated with the University of Patras to translate a synthetic biology comic book into Twi (Ghanaian language), enhancing synthetic biology education/awareness for middle school students in Ghana.
  - Co-authored an article on safe gold prospecting in Ghana, published in an interdisciplinary journal featuring research projects from various IGEM teams, compiled by Stony Brook University.
  - Led the development of the team's wiki, creating an engaging website to showcase the project's scientific narratives and research findings.
- **Impact**:
  - Contributed to winning the Safety and Biosecurity Grant ($5000) from the iGEM Foundation, one of only five teams selected out of forty-eight.
  - Promoted science literacy through language localization and publication, making synthetic biology concepts accessible to both academic and general audiences.
  - Published research findings in an interdisciplinary journal.

  Principal Investigator: Dr. Elena Rosca

Papers

Publication: Joseph A. Mensah, Justice K. Appati, **Elijah K.A Boateng**, Eric Ocran, and Louis Asiedu, "FaceNet recognition algorithm subject to multiple constraints: Assessment of the performance," *Scientific African*, vol. 23, p. e02007, 2024. doi: 10.1016/j.sciaf.2023.e02007.

Thesis: **Elijah K.A Boateng**, "Face Recognition under Multiple Constraints with Application to Attendance Management," Ashesi University, 2023. [Online]. Available: https://hdl.handle.net/20.500.11988/1029. **(Best Computer Science Thesis)**

Article: **AshesiGhana team**, "An Affordable & Fast Approach to Gold Prospecting in Ghana," AshesiGhana, 2022. [Online]. Available: https://2022.igem.wiki/stony-brook/collaborations/global-journal.

Technical Skills & Research Interests

- **Languages**: Python, Java, C++, SQL, Basic web technologies(HTML, CSS, JavaScript)
- **Tools**: Docker, Git
- **Platforms**: Linux, Windows
- **Research Interests**: Multi-agent systems, AI for Low Resource Languages, AI for Healthcare, Forensics and Environmental Safety, Machine Learning, Algorithms and Complexity

Work Experience

Ashesi University Eastern Region, Berekuso
Teaching Assistant – OOP(Java) & Data Structures and Algorithms & Pre-Calculus *Aug 2023 – Aug 2024*

- **Organized tutorial sessions for students**: Reinforced course concepts and enhanced their understanding of the material through hands-on activities for topics in programming and mathematics.
- **Assessed and graded assignments, quizzes, and problem sets**: Provided constructive feedback to help students improve their Performance.
- **Collaborated with the course instructor**: Developed and implemented effective teaching strategies, tailoring support to students needing more class attention.

Black Star Brokerage Greater Accra, Accra
Intern, Summer Analyst *May 2021 - Aug 2021*

- **Built and deployed a research portal**: Developed a portal for the business's brokerage side, which has traded two and a half billion Ghana cedis worth of bonds to analyze secondary and primary market bond transactions in terms of yield and volumes traded. This enabled the business to monitor government fiscal targets, the percentage met on the primary market, bonds trading, and price movements on the secondary market.
- **Data cleaning and visualization**: Cleaned and visualized large volumes of data to ensure accuracy and clarity, supporting informed decision-making and analysis.

Black Star Advisors Greater Accra, Accra
Intern, Software Engineer *May 2021 - Aug 2021*

- **Built an analytics system**: Collaborated with a co-intern to build a performance, risk, and benchmarking attribution system using Python Streamlit for pension and mutual funds for the asset management division, which has a hundred million dollars in AUM (Assets Under Management).
- **Performance and risk comparison**: Enabled the organization to compare the Performance and associated risk statistics of more than twenty portfolios to each other and against a benchmark.

Ashesi Science Engineering Entrepreneurship Design (SEED) Journal Eastern Region, Berekuso
Backend Developer *Mar 2020 – Nov 2020*

- **Developed a secure authentication framework**: Created a secure authentication framework for website login and signup using Python Flask.
- **Validation stream for submitted papers**: Designed and implemented a validation stream for submitted papers, integrating MySQL and Python Flask for efficient data management.
- **Role and permission management**: Implemented role and permission management using Python Flask and JavaScript, ensuring secure and customized user access for the website.

Personal Projects
Developer *Mar 2021, Jun 2022*

- **Developed a GPA forecasting tool**: Created a Python-based GPA forecasting tool in 2022, enabling students to project required grades to achieve specific academic targets on a semester and yearly basis or at the end of their degree. The institution adopted the tool for academic planning and has benefited more than 1000 students since deployment.
- **Implemented a peer tutor booking system**: Designed and deployed a campus-wide peer tutor booking system in 2021, allowing proficient students to register as tutors and enabling others to book tutoring sessions, with over 100 tutors registered across all majors and about 550 booked sessions.

Leadership & Volunteer Experience

Ashesi Student Council Eastern Region, Berekuso
Academic Chairperson *Feb 2022 – Mar 2023*

- **Academic Council Member**: Actively contributed to key decisions and policies that impacted the academic direction of the institution.
- **Student Representative**: Participated in the Ashesi Judicial Council, ensuring fair representation and adherence to established rules and regulations. Met with the Provost to discuss students' academic challenges and how we could mitigate them.

Ashesi Math Resource Centre Eastern Region, Berekuso
Lead *Jan 2021 – Aug 2022*

- **Coordinated Math Centre Activities**: Matched over 20 tutors with about 75 students for targeted assistance in diverse math classes.
- **One-on-One Sessions**: Assisted fellow 33 students facing challenges in mathematics, ensuring a deeper understanding of the subject matter, which helped about 25 of them move at least one letter grade up.
- **Encouraged the Study of Mathematics**: Promoted peer-to-peer support and campaigns, such as the PI Day initiative, to encourage the study of mathematics.

Berekuso Math Project Eastern Region, Berekuso Community School
Volunteer Tutor *Sep 2020 – Dec 2021*

- **Tutored Mathematics**: Volunteered to tutor over 100 students in preparation for their high school entrance exams, resulting in a significant improvement in the mathematics pass rate.